LIFTING A SYNTOMIC, SMOOTH OR ETALE MORPHISM TO A CLOSED SUBSCHEME UP TO A ZARISKI COVERING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the three types of morphism, all handled by the same reduction and lifting step.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Closed immersion and the structure of a syntomic, smooth or etale morphism over a nilpotent thickening. Same chapter together with the chapter Morphisms, the description of liftings of syntomic, smooth and etale morphisms along a closed immersion. Supplies the local lifting used to build the morphisms V_i.
- Zariski coverings and locality of the lifting problem. Same chapter, the locality of the assertion in the source U and in the base Y. Supplies the reduction to an affine local situation.

Source reference. Original lemma 00T0: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6873–6873. Navigation pointer invoked at proof line 6873; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04B1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6880–6880. Navigation pointer invoked at proof line 6880; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04D1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6882–6882. Navigation pointer invoked at proof line 6882; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : X \to Y$ be a closed immersion of schemes. Let $U \to X$ be a syntomic (resp. smooth, resp. étale) morphism. Then there exist syntomic (resp. smooth, resp. étale) morphisms $V_i \to Y$ and morphisms $V_i \times_Y X \to U$ such that $\{V_i \times_Y X \to U\}$ is a Zariski covering of U.*

Proof. Let us prove the lemma when $\tau = \textit{syntomic}$. The question is local on U. Thus we may assume that U is an affine scheme mapping into an affine of Y. Hence we reduce to proving the following case: $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(A/I)$, and $U = \mathrm{Spec}(\overline{B})$, where $A/I \to \overline{B}$ be a syntomic ring map. By Algebra, Lemma 00T0 we can find elements $\overline{g}_i \in \overline{B}$ such that $\overline{B}_{\overline{g}_i} = A_i/IA_i$ for certain syntomic ring maps $A \to A_i$. This proves the lemma in the syntomic case. The proof of the smooth case is the same except it uses Algebra, Lemma 04B1. In the étale case use Algebra, Lemma 04D1. □